

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term Examination, 2020
Department of Materials Science and Engineering

MSE 1101
(Introduction to Materials Science and Engineering)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Explain why hydrogen fluoride (HF) has a higher boiling temperature than hydrogen chloride (HCl) (19.4 °C vs -85.0 °C), even though HF has a lower molecular weight. (05)
(b) What are the four components of the discipline of materials science and engineering and how they are interrelated? (10)
(c) The atomic radii of Mg^{2+} and F^- ions are 0.072 and 0.133 nm, respectively. (15)
(i) Calculate the force of attraction between these two ions at their equilibrium interionic separation (i.e., when the ions just touch one another).
(ii) What is the force of repulsion at this same separation distance?
- Q 2. (a) Define lattice points and unit cell. Calculate the atomic packing factor (APF) of FCC and BCC crystals. (20)
(b) Explain polymorphism and allotropy with suitable examples. (10)
- Q 3. (a) Draw a typical stress-strain diagram for a tensile test on mild steel. Discuss the information that may be obtained from such a test. (15)
(b) Define hardness. Discuss the Knoop and Vickers Microindentation hardness tests and describe the precautions which must be taken in its operation. (15)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) What do you mean by composite materials? Discuss the functions of matrix and reinforcements in composite materials with proper example. (08)
(b) Explain with detail sketches why length and diameter of the fiber materials play the vital roles for the enhancement of load bearing capability of composite materials. (12)
(c) A continuous aligned glass fiber-reinforced composite consists of 30% volume of glass fibers having an elastic modulus of 65 GPa and 60% volume of an epoxy resin that, when hardened, has an elastic modulus of 4 GPa. If the cross-sectional area of the composite is 250 mm² and a stress of 50 MPa is applied in the direction of fiber alignment then, (10)
(i) Compute the elastic modulus of the composite in longitudinal and transverse direction.
(ii) Determine the strain in each phase of composite and also the stress in each phase of composite.
- Q 5. (a) What do you mean by intrinsic semiconductor? Briefly discuss the effect of temperature on the conductivity of extrinsic semiconductor. (10)
(b) What is luminescence? Briefly discuss the mechanism of LED with a neat sketch. (12)
(c) Derive the conditions of band gap energies for non-metals to be transparent, opaque and colored. (08)
- Q 6. (a) Define the following terms: (05)
(i) Magnetic dipole moment
(ii) Magnetic domain
(b) What is NDT? If you have a plastic pipe and an aluminum pipe, then what non-destructive tests will you carry out for the inspection of surface and internal defects of the samples? (10)
(c) What is meant by magnetic susceptibility? Discuss the effect of temperature on magnetism for different magnetic materials with necessary figures. (15)

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Ph 1127
(Optics and Waves)

Time: 1.5 hours

Full Marks: 120

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(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Two simple harmonic vibration acting at right angles to each other have the time period in the ratio 1:2. The phase difference between the two vibration is $\pi/2$. Show graphically that the resultant curve is parabola. (12)
- (b) Explain clearly the terms: free vibration, forced vibration and resonance. Give example of each. (10)
- (c) Find whether the discharge of a condenser for the following inductive circuit is oscillating $c = 0.1 \mu\text{F}$, $L = 10 \text{ mH}$, $R = 200 \text{ Ohms}$. If the circuit is oscillatory, calculate its frequency. (08)
- Q 2. (a) Investigate theoretically the formation of stationary waves. Show that in such waves the net transmission of energy is zero. (12)
- (b) What is meant by ultrasonic waves? Mention the properties of ultrasonic waves. (10)
- (c) A person is standing on a platform. A railway engine moving away from the person with a speed of 80 Km/hr blow a whistle of pitch 650 Hz. Calculate the apparent pitch of the whistle as heard by the person. Velocity of sound in air = 340 m/s. (08)
- Q 3. (a) Explain what is meant by “bel” and “phon”. Discuss the factors influencing Loudness. (10)
- (b) Define the intensity and energy density at a point in a plane wave of sound. Obtain an expression for the intensity of a plane wave in terms of acoustic pressure. (10)
- (c) What are the acoustic requirements of a good auditorium? How an auditorium can be used for speech, music and opera. (10)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) What is meant by interference of light? Discuss the conditions for interference. (08)
- (b) Give a diagram showing clearly how coherent sources are produced in a biprism. Derive the formula for the fringe width in the biprism experiment. (14)
- (c) Newton's rings are observed in reflected light of wavelength 5.9×10^{-5} cm. The diameter of the 10th dark ring is 0.4 cm. Find the radius of curvature of the lens. (08)
- Q 5. (a) What are the advantages of a compound microscope over a simple microscope? Calculate its magnifying power. (13)
- (b) Explain the use of diffraction grating in measuring wavelength of light. (10)
- (c) Deduce the missing orders for a double slit Fraunhofer diffraction pattern, if the slit widths are 0.20 mm and they are 0.1 mm apart. (07)
- Q 6. (a) How can you say that light wave is a transverse wave? Argue your answer. (07)
- (b) How can (i) elliptically polarized light and (ii) circularly polarized light be produced? How would you distinguish between un-polarized light and circularly polarized light? (15)
- (c) Determine the specific rotation of the given sample of sugar solution if the plane of polarization is turned through 13.2° . The length of the tube containing 10% sugar solution is 20 cm. (08)

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Ch 1127
(Inorganic and Physical Chemistry)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Use molecular orbital theory to compare the relative stabilities of F_2 and F_2^+ . (10)
(b) Explain different types of H-bonding. Explain O-nitrophenol has low melting point and boiling point as compared to m- and p-nitrophenol. (10)
(c) Calculate the second electron gain enthalpy of O_2 from the following information for Mg, O_2 and MgO : (10)
For Mg(s), $\Delta H_{\text{sub}} = 148 \text{ kJ/mol}$
1st ionization energy for (Mg) = 738 kJ/mol
2nd ionization energy for (Mg) = 1450 kJ/mol
Bond dissociation for $O_2 = 499 \text{ kJ/mol}$
1st electron gain enthalpy for $O_2 = -141 \text{ kJ/mol}$
For MgO, Lattice energy = 3890 kJ/mol
For MgO, Enthalpy of formation = -602 kJ/mol
- Q 2. (a) What is salt-bridge? Why is it used? Explain its working principle giving an example. (08)
(b) Define EMF. Derive Nernst's equation for the determination of EMF of a cell. (10)
(c) With the help of a diagram describe the effect of humidity and dust particle on the rate of atmospheric corrosion. (12)
- Q 3. (a) What is activation energy? What role does activation energy play in chemical kinetics? (10)
(b) What is the half-life of a compound if 75 % of a given sample of the compound decomposes in 60 min? Assume first-order kinetics. (10)
(c) Potassium dichromate oxidizes potassium iodide in acid solution liberating iodine use half reaction method to complete and balance this reaction. (10)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) State the phase rule? For the following reaction (10)
$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

How many number of phases, components and degrees of freedom are available in this equilibrium?
(b) Draw a neat phase diagram of one component water system and explain the curves, areas and point with reference to the phase rule. (12)
(c) An alloy AB contains 40 % A in 1Kg. AB on cooling forms eutectic containing 60 % A and 40 % B and the rest of B. What is the amount of B formed during eutectic formation? (08)
- Q 5. (a) What are emulsions? Discuss the role played by an emulsifier in the process of emulsification. (10)
(b) What is electrical double layer? How does thermodynamic potential differ from Zeta potential? (10)
(c) What is micelle? How will you detect a colloidal solution forms micelle? (10)
- Q 6. (a) Briefly explain how would you remove electrolyte and impurities from colloidal solution. (12)
(b) Write down the uses and limitations of phase rule. (08)
(c) Classify the colloids into different groups with special reference to their affinity towards the dispersion medium. (10)

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EEE 1127
(Electrical Engineering Fundamentals)

Time: 1.5 hours

Full Marks: 120

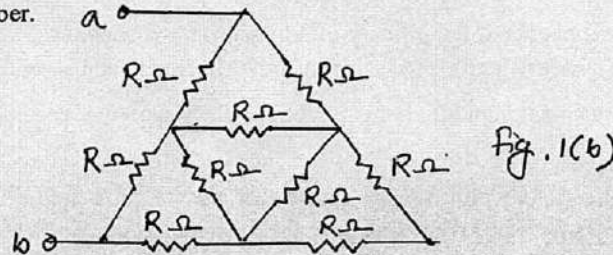
- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
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Section A

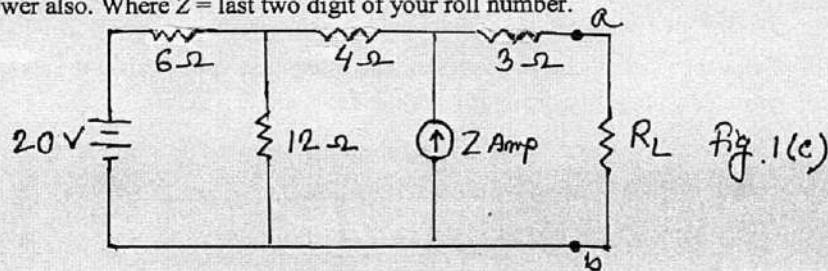
(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Define instruments and classify them. Explain self and mutual inductance. (10)

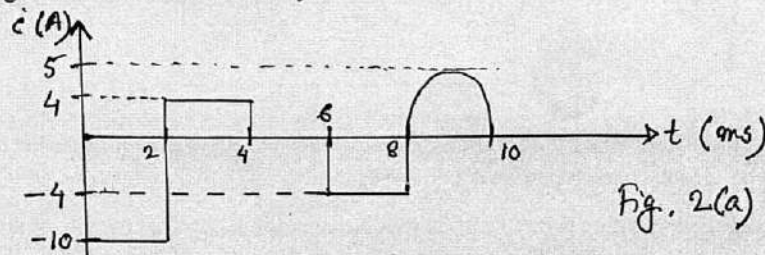
- (b) Obtain the equivalent resistance R_{ab} for the following circuit given Fig. Q.1(b), where $R = \text{last digit of your roll number}$. (10)



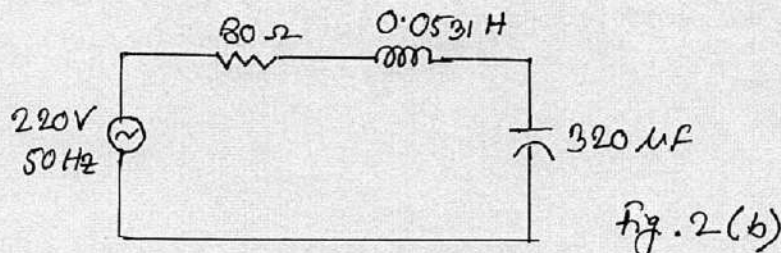
- (c) Find the value of R_L for maximum power transfer of the circuit given in Fig. Q.1(c). Find the maximum power also. Where $Z = \text{last two digit of your roll number}$. (10)



- Q 2. (a) Define form factor, r.m.s value. Find the (i) average value (ii) r.m.s values (iii) form factor of the following waveforms over one full cycle. (10)



- (b) Why sinusoidal wave is widely used? Calculate real power, reactive power, apparent power, power factor of the circuit shown in Fig. Q. 2(b). (10)



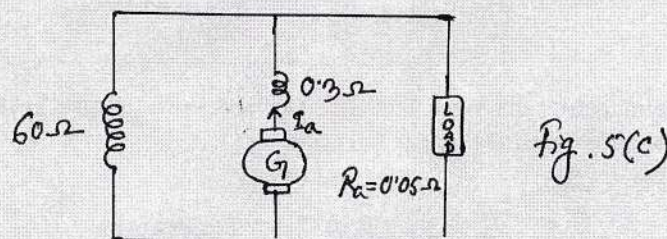
- (c) Find the resonance condition of a series RLC circuit. Hence, show that the frequency response at resonance condition is inversely proportional to the square root of capacitance. (10)

- Q 3. (a) Define (i) Residual magnetism (ii) Coercive force (iii) Magnetism saturation (iv) Hysteresis. (08)
- (b) Write down the differences between magnetic circuit and DC circuit. Hence, show that "Ampere's circuital law in magnetic circuit is equivalent of Kirchoff's voltage law in DC circuit". (12)
- (c) Write down the advantages and disadvantages of PMMC instruments for measuring devices. (10)
- Derive the expression of multiplication factor, $M = 1 + \frac{R_m}{R_{sh}}$ for range extension of a PMMC Ammeter.

Section B

(Answer ANY TWO questions from this section in answer script B)

- Q 4. (a) What will happen if the primary of a transformer is supplied with a DC supply? Show that emf per turn ($emf/turn$) is the same in both the primary and secondary windings. (10)
- (b) Derive the maximum efficiency condition of a transformer considering core loss and copper losses. (10)
- (c) In a 25 KVA, 1000/400 V, 1- ϕ transformer, the core loss and full load copper losses are 200 W and 300 W respectively. Calculate the efficiency at unity power factor on full load and half full-load. (10)
- Q 5. (a) Differentiate between lap winding and wave winding. "The voltage generated in armature in AC but the voltage collected in DC"-Justify the statement. (10)
- (b) Draw the external characteristics of a dc shunt generator. Explain the reasons for the drop in terminal voltage of a shunt generator under load. (10)
- (c) Calculate the generated emf for the following DC machine as in Fig. Q. 5 (c). (10)



- Q 6. (a) What is back emf of a dc motor? What are the factors controlling motor speed? Draw the speed-torque characteristics of a dc series motor. (10)
- (b) Explain the working principle of an induction motor. Show that the slip of an induction motor is equal to 1 at stationary. (10)
- (c) Briefly explain the operation of an alternator. A 440 V, shunt motor has armature resistance of 0.8Ω and field resistance of 200Ω . Determine the back emf when giving an output of 7460 watt at 85 percent efficiency. (10)

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~~Math~~
~~MSE~~ 1127

(Differential and Integral Calculus)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
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Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) A function $f(x)$ is defined by (15)

$$f(x) = \begin{cases} 3x + 4; & x \geq 1 \\ x + 6; & -\frac{5}{2} \leq x < 1 \\ -3x - 4; & x < -\frac{5}{2} \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x = -\frac{5}{2}$

- (b) If $y = \frac{2x+1}{x^2-3x+2}$, then find y_n . (10)

- (c) Calculate the value of $\sin 46^\circ$ using the value of $\sin 45^\circ$. (05)

- Q 2. (a) Is Rolle's theorem applicable for the function $f(x) = \frac{3}{2-x^2}$ in $[-2, 2]$? Justify your answer. (05)

- (b) Verify Euler's theorem for the function $u = x^3 \ln\left(\frac{y}{x}\right)$. (10)

- (c) Find all the possible asymptotes of the curve (15)

$$x^2y^2 - 3x^2y - 3xy^2 + 2x^2 - 4y^2 - 3x - 4y = 0$$

- Q 3. (a) If $F(v^2 - x^2, v^2 - y^2, v^2 - z^2) = 0$ where v is a function of x, y, z show that (15)

$$\frac{1}{x} \frac{\partial v}{\partial x} + \frac{1}{y} \frac{\partial v}{\partial y} + \frac{1}{z} \frac{\partial v}{\partial z} = \frac{1}{v}$$

- (b) Find the circle of curvature at $(0, 1)$ of the curve $y = x^3 + 2x^2 + x + 1$ (15)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. Integrate the followings :

(a) $\int \frac{e^x - 1}{e^{x+2}} dx$ (08)

(b) $\int \frac{dx}{(2x+3)\sqrt{x^2+3x+2}}$ (11)

(c) $\int \frac{dx}{(x-a)\sqrt{(x-a)(b-x)}}$, when $a < x < b$ (11)

- Q 5. (a) Evaluate i) $\int_0^\pi \frac{x dx}{(a^2 \cos^2 x + b^2 \sin^2 x)^2}$ (10+08)

ii) $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \frac{n^2}{(n+3)^3} + \dots + \frac{1}{8n} \right]$

- (b) If $I_n = \int_0^{\frac{\pi}{2}} \theta \sin^n \theta d\theta$ and $n > 1$ then prove that $I_n = \frac{n-1}{n} I_{n-2} + \frac{1}{n^2}$ (12)

- Q 6. (a) Define Gamma and Beta function. Find the relation between Gamma and Beta function. (15)

- (b) Evaluate $\iiint_G xyz dv$, where G is the solid in the first octant is bounded by the parabolic cylinder $z = 2 - x^2$ and the planes $z = 0$, $y = x$, $y = 0$. (15)